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Bachelor of Science Thesis

*Embedded Implementation and Comparison of Attitude Estimation
Methods*

*Implementacja na urządzenie wbudowane algorytmów odtwarzania orientacji i ich
porównanie*

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1. Abstract

The main goal of this thesis is to compare various attitude estimation algorithms (with a North unreferenced Yaw) using data from a 6-degree-of-freedom MEMS inertial measurement unit (6DOF IMU). The comparison will be made based on two main criteria, estimation accuracy and computation cost.

The experimental evaluation will include following algorithms:

- Complementary filter
- Mahony filter
- Extended Kalman Filter (EKF)
- Madgwick filter

The estimation accuracy will be measured under different conditions and at different IMU sampling frequencies. This data will be interpreted using the mentioned algorithms and compared with a reference attitude measurement (obtained by a reference sensor or a motion capture system). Test scenarios will cover measurements in both static conditions (with low external interference) and dynamic conditions (with linear or centrifugal accelerations added). The tests should enable comparison of how different algorithms handle noise directly from sensors and how those filters react to external noise, introduced by the environment.

To compare the computational cost, selected algorithms will be implemented and benchmarked on an STM32 family microcontroller in the C programming language. This will enable a direct measurement of time or memory overhead introduced by the filter.

1.1. Introduction

2. Methods of Attitude Representation

An attitude of a body is a description of angular position of its body (sensor) frame, relative to the world (Earth) reference frame.

The most commonly known method for expressing the attitude of a body is through the use of Euler angles. This method describes the attitude as a sequence of three rotations around three separate axis of reference. While this method is simple and very intuitive it is also, arguably the most problematic of all methods mentioned here. The biggest problem of Euler angles representation is the phenomenon of gimbal lock. It appears when two rotation axis become parallel, which results in the loss of one degree of freedom. This can lead to singularities in the mathematical representation, which in turn causes difficulties and discontinuities in attitude representation. On top of that, calculating rotations of vectors between two frames described with Euler angles involves the use of, computationally expensive trigonometric functions. Additionally, due to the fact that rotations around each axis are calculated separately, the order of rotations matters, and gives different result based on chosen convention.

Most of the problems of Euler angles can be solved by using rotation matrices. A rotation matrix is a 3×3 orthogonal matrix with determinant equal to 1, which represents a rotation in three-dimensional space. This method does not suffer from gimbal lock and provides a unique way of representing rotations, at a cost of reducing intuitiveness of the representation. Rotating a vector using a rotation matrix requires only matrix multiplication. The biggest issue with rotation matrices is their memory footprint. They require 9 parameters to represent a rotation, which is more than necessary and leads to additional numeric complexity and instability. The matrixes will also accumulate numerical errors during rotation concatenation, and will require re-orthogonalization to maintain their properties.

Because of that one more improvement can be made, when it comes to attitude representation, by using unit quaternions. Quaternions are a mathematical construct that extends complex numbers and can be used to represent rotations in three-dimensional space. It consists of four components: one real (scalar) part, and three imaginary parts forming a 3d vector. Quaternions bring the same improvements as rotation matrices, while also reducing the memory footprint to less than half. They also reduce the number of multiplications required for vector rotation and concatenation, which makes them a natural choice for representing rotations. Quaternions still suffer from the same numerical issues as rotation matrixes but maintaining the unit norm requires much less computational effort.

2.1. Representing Rotations as Quaternions

Quaternions are defined as a four-dimensional extension of complex numbers, consisting of one real part and three imaginary parts. A quaternion can be represented as:

$$\mathbf{q} = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} = [q_0 \ q_1 \ q_2 \ q_3]^T, \quad \mathbf{q} \in \mathbb{H} \quad (2.1)$$

where a, b, c, d are real numbers and i, j, k are the basis vectors. The set of all quaternions is denoted by $\mathbb{H}$, for William Rowan Hamilton, who first introduced quaternions in 1843. Neutral element of this set is the unit quaternion:

$$\mathbf{q}_I = 1 + 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = [1 \ 0 \ 0 \ 0]^T \quad (2.2)$$

Other common notation for quaternions is to represent them as a scalar part and a vector part:

$$\mathbf{q} = [s, \mathbf{v}] \in \mathbb{R} \times \mathbb{R}^3, \quad s = q_0, \mathbf{v} = [q_1 \ q_2 \ q_3]^T \quad (2.3)$$

Multiplication between quaternions, called a Hamilton product is defined as:

$$\mathbf{q}_1 \otimes \mathbf{q}_2 = \begin{bmatrix} s_1 s_2 - \mathbf{v}_1^T \mathbf{v}_2 \\ s_1 \mathbf{v}_2 + s_2 \mathbf{v}_1 + \mathbf{v}_1 \times \mathbf{v}_2 \end{bmatrix} \quad (2.4)$$

Quaternion conjugate is defined as:

$$\mathbf{q}^* = \begin{bmatrix} s \\ -\mathbf{v} \end{bmatrix} \quad (2.5)$$

For a quaternion $\mathbf{q}$ to represent a valid rotation, it must have a unit norm:

$$\|\mathbf{q}\| = \sqrt{\mathbf{q}^* \otimes \mathbf{q}} = \sqrt{\mathbf{q} \otimes \mathbf{q}^*} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} = 1 \quad (2.6)$$

Let a quaternion representing a rotation of frame A with respect to frame B be denoted as ${}^B_A \mathbf{q}$, and a vector written in frame B be denoted as ${}^B \mathbf{v}$, with $\mathbf{u}$ being a unit vector. Any rotation θ around a unit vector ${}^B \mathbf{u}$ can be represented as a quaternion:

$${}^B_A \mathbf{q} = \begin{pmatrix} \cos \frac{\theta}{2} \\ {}^B \mathbf{u} \sin \frac{\theta}{2} \end{pmatrix} \quad (2.7)$$

Every time a orientation is described using quaternion without specifying the frames, it will be assumed that the quaternion describes the rotation of world frame with respect to the body frame.

$$\mathbf{q} = {}^B_W \mathbf{q} \quad (2.8)$$

A rotation described by a quaternion is unique, up to its sign. Both $\mathbf{q}$ and $-\mathbf{q}$ represent the same rotation in space. To inverse the rotation described by a quaternion, the quaternion conjugate should be used:

$${}^A_B \mathbf{q} = {}^B_A \mathbf{q}^* \quad (2.9)$$

Quaternion rotations can be easily concatenated together, using Hamilton product:

$${}^C_A \mathbf{q} = {}^B_A \mathbf{q} \otimes {}^C_B \mathbf{q} \quad (2.10)$$

Given two quaternions $\mathbf{p}$ and $\mathbf{q}$, the cosine of the angle Ω subtended by the arc between them is equal to the dot product of the two quaternions:

$$\cos \Omega = \langle \mathbf{p}, \mathbf{q} \rangle = s_1 s_2 + \mathbf{v}_1^T \mathbf{v}_2 \quad (2.11)$$

Rotation quaternions can be interpolated using spherical linear interpolation (SLERP):

$$SLERP(\mathbf{q}, \mathbf{p}, \alpha) = \frac{\sin(\alpha\Omega)}{\sin(\Omega)} \mathbf{p} + \frac{\sin([1 - \alpha]\Omega)}{\sin(\Omega)} \mathbf{q}, \alpha \in [0, 1] \quad (2.12)$$

This interpolation can be also approximated with a simple linear interpolation (LERP), reducing the computational cost at the cost of the accuracy and loss of the unit norm.

$$LERP(\mathbf{q}, \mathbf{p}, \alpha) = (1 - \alpha)\mathbf{q} + \alpha\mathbf{p}, \alpha \in [0, 1] \quad (2.13)$$

To perform any mathematical operation between a vector and quaternion, it is a common practice to first convert the vector into a pure quaternion $\mathbf{p}$ i.e. quaternion with a zero scalar part. The transformed vector can be later extracted from the pure quaternion by taking its vector part.

$${}^B \mathbf{p} = \begin{bmatrix} 0 \\ {}^B \mathbf{v} \end{bmatrix} \quad (2.14)$$

To perform a rotation of a vector ${}^B \mathbf{v}$ from frame B to frame A, the following operation can be performed:

$${}^A \mathbf{p} = {}^B_A \mathbf{q} \otimes {}^B \mathbf{p} \otimes {}^B_A \mathbf{q}^* \quad (2.15)$$

Quaternion can also be converted into a rotation matrix:

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} 1 - 2(c^2 + d^2) & 2(bc - ad) & 2(bd + ac) \\ 2(bc + ad) & 1 - 2(b^2 + d^2) & 2(cd - ab) \\ 2(bd - ac) & 2(cd + ab) & 1 - 2(b^2 + c^2) \end{bmatrix} \quad (2.16)$$

2.2. Chosen frame of reference

The global frame of reference will be represented using NED (North-East-Down) coordinate system. The NED system is a right-handed local tangent plane coordinate system. The x-axis of this system points towards the north, the y-axis points towards the east and the z-axis points downwards, aligning with local gravity vector.

Because of common quaternion conventions, every quaternion representing a rotation will describe a orientation of the world frame relative to the body frame. The body frame will be denoted as $\{B\}$, the world frame as $\{W\}$. Additionally, any estimator will provide only an approximation of the real world frame rotation, this can be fought of as a separate estimator frame $\{E\}$.

3. Attitude from IMU Measurements

A typical MEMS IMU consists of 2 separate sensors. A rate gyroscope, measuring angular velocity along 3 sensor frame axis, in rad/s . As well as a 3-axis accelerometer, measuring linear acceleration along the same three axis, in m/s^2 . Neither one of those two sensors can be used alone to provide an reliable attitude observation in real conditions.

3.1. Attitude from Accelerometer

Measurement model for accelerometer observations a_y can be represented as:

$$\mathbf{a}_y = {}^W_B \mathbf{R}(\dot{\mathbf{v}} + \mathbf{g}) + \mathbf{b}_a + \boldsymbol{\mu}_a \quad (3.1)$$

Where:

${}^W_B \mathbf{R}$ – Rotations matrix representing transformation from world to body frame

$\mathbf{g}$ – Gravity vector in word frame

$\dot{\mathbf{v}}$ – Body frame acceleration

$\mathbf{b}_a$ – Bias of the accelerometer

$\boldsymbol{\mu}_a$ – Measurement noise, including all unmodeled dynamics of the system

Assuming perfectly static conditions and 0 bias term, the accelerometer measures only the gravitational acceleration vector in the sensor frame. This measurement can be used to calculate a transformation aligning the body frame to the NED frame down direction. It is important to note that this method doesn't provide full orientation of the body, only the attitude can be estimated, while the heading (yaw) remains unobservable.

In most common cases, the sensor will be subject to additional, and often unpredictable external forces, which will make accelerometer only attitude estimation unreliable or even dangerous when used for control purposes.

3.2. Attitude from Gyroscope

Gyroscope measurements Ω_y can be represented as:

$$\Omega_y = {}^B\Omega + \mathbf{b}_g + \mu_g \quad (3.2)$$

Where:

${}^B\Omega$ – Real angular velocity vector in the sensor frame

$\mathbf{b}_g$ – Gyroscope bias, slowly varying over time

μ_g – Measurement noise, including all unmodeled dynamics of the system

Assuming an initial orientation, which cannot be measured by the gyroscope, the orientation can be estimated by integrating the angular velocity over time. This method gives an full orientation estimate, which is mostly unaffected by external linear accelerations. However, due to measurement noise, numerical integration errors and most importantly, the slowly varying bias, this estimate will drift over time, making it unreliable for long term use.

3.3. Attitude from Sensor Fusion

While achieving a reliable and universal attitude estimation from either of the two sensors alone is not possible, combining the measurements from both sensors can provide a much more accurate and stable estimate. The gyroscope measurement is much more reliable and accurate in short term observations, so its going to serve a the "base" of the algorithm. The accelerometer measurement will correct the errors that accumulate in the gyroscope based estimate, as well as providing an initial orientation estimate.

4. Chosen Attitude Estimation Algorithms

4.1. Mahony filter

Mahony filter is a name commonly used to refer to a group of complementary filters derived by Robert Mahony in [1]. The main idea behind those algorithms is to provide a computationally inexpensive solution for estimating attitude from cheap MEMS IMU sensors. Mahony expands on the idea of the classic complementary filter, by introducing a feedback term to the gyroscope measurement, which is an integral of the error between the accelerometer measurement and the expected gravity vector in the sensor frame. This feedback term helps to correct the drift in the gyroscope based estimate, while still maintaining the computational simplicity of a complementary filter. In this work, the *explicit complementary filter* version of the algorithm proposed in [1] will be used.

To derive the correction term for the filter, Madgwick uses an error that is the relative rotation from body frame to the estimator frame. In practice, the real body frame rotation is unknown, so an estimated rotation derived from available measurements R_y is used instead.

$$E_{tr} = \frac{1}{2} \text{tr}(\mathbf{I}_3 - \hat{\mathbf{R}}^T \mathbf{R}_y) = 1 - \cos(\theta) \quad (4.1)$$

Where θ is the angle associated with the rotation between those frames. The filter dynamics, in quaternion form are given by:

$$\begin{aligned} \dot{\hat{\mathbf{q}}} &= \frac{1}{2} \hat{\mathbf{q}} \otimes \mathbf{p}(\boldsymbol{\Omega}_y - \hat{\mathbf{b}} + k_p \boldsymbol{\omega}) \\ \dot{\hat{\mathbf{b}}} &= -k_I \boldsymbol{\omega} \end{aligned} \quad (4.2)$$

The goal of the estimator will be to find a term $\boldsymbol{\omega}$ that will drive this error to 0. The correction term introduced by Madgwick is:

$$\boldsymbol{\omega} = \text{vec}(\mathbb{P}_a(\hat{\mathbf{R}}^T \mathbf{R}_y)) = \sin(\theta) \mathbf{u} \quad (4.3)$$

With the normalized accelerometer measurement v and the expected gravity vector $\hat{v} = \mathbf{R}(\hat{\mathbf{q}})^T \mathbf{g}$, $g = [0, 0, 1]^T$, the correction term can be expressed as:

$$\boldsymbol{\omega} = v \times \hat{v} \quad (4.4)$$

With the correction term defined, the full explicit complementary filter dynamics can be implemented as follows:

$$\begin{aligned}\dot{\hat{q}} &= \frac{1}{2} \hat{q} \otimes p(\Omega_y - \hat{b} + k_p \omega) \\ \dot{\hat{b}} &= -k_I \omega \\ \omega &= k_I (v \times \hat{v})\end{aligned}\tag{4.5}$$

The estimator can be parameterized by two gains, k_p and k_I . The k_p parameter controls the instantaneous filter correction. If its too high, the filter will be very sensitive to accelerometer noise, while if its too low, the filter will be slow to correct for gyroscope drift, or even diverge all together. The k_I parameter controls the integral correction term, which is responsible for estimating the gyroscope bias.

4.2. AQUA Complementary filter

In [2] Roberto G. Valenti *et. al.* describes a orientation estimation algorithm, which is an interesting variation on an standard complementary filter algorithm. The main idea behind his approach is to provide estimation as the algebraic solution of a system from inertial and optionally magnetic observations. During the correction step, the algorithm finds the tilt quaternion from accelerometer and the heading quaternion from magnetometer observations separately. This allows to avoid the problem of magnetic distortion affecting the attitude estimation.

The dynamics of the filter prediction step, as proposed in [2], but adjusted for notation conventions are given by:

$${}^B_E \dot{\hat{q}}_{\omega, t_k} = \frac{1}{2} \hat{q}_{\omega, t_{k-1}} \otimes p(\Omega_{y, t_k})\tag{4.6}$$

This equation can be computed numerically, using a simple Euler integration method:

$${}^B_E \hat{q}_{\omega, t_k} = \hat{q}_{\omega, t_{k-1}} + \dot{\hat{q}}_{\omega, t_k} \Delta t\tag{4.7}$$

In the correction step, an delta quaternion is computed from the accelerometer observation, which is then used to correct the predicted orientation:

$$\hat{q}_{t_k} = {}^E_W \widehat{\Delta q}_{acc} \otimes {}^B_E \hat{q}_{\omega, t_k}\tag{4.8}$$

The delta quaternion Δq_{acc} satisfies the following system of equations:

$$\begin{cases} 2(\Delta q_{1, acc} \Delta q_{3, acc} + \Delta q_{0, acc} \Delta q_{2, acc}) & = G_x \\ 2(\Delta q_{2, acc} \Delta q_{3, acc} + \Delta q_{0, acc} \Delta q_{1, acc}) & = G_y \\ \Delta q_{0, acc}^2 - \Delta q_{1, acc}^2 - \Delta q_{2, acc}^2 + \Delta q_{3, acc}^2 & = G_z \end{cases}\tag{4.9}$$

Where $G = R_W^B(\hat{q})g = [G_x, G_y, G_z]^T$, $g = [0, 0, 1]^T$, is the gravity vector predicted from the orientation estimate. After defining the component $\Delta q_{3, acc} = 0$ or $\Delta q_{2, acc} = 0$, the solution to this system of can be found, one for each case. Two solutions were given to avoid singularities in the solution, depending on the value of G_z :

$$\Delta q_{acc} = \begin{cases} [\sqrt{\frac{G_z+1}{2}} & \frac{G_y}{2G_z+1} & -\frac{G_x}{2G_z+1} & 0] & , G_z >= 0 \\ [-\frac{G_y}{2(1-G_z)} & -\sqrt{\frac{1-G_z}{2}} & 0 & -\frac{G_x}{2(1-G_z)}] & , G_z < 0 \end{cases}\tag{4.10}$$

Applying the full correction term will instantaneously rotates the predicted orientation to match the tilt observed by the accelerometer. This is undesirable, as the accelerometer measures both the gravity vector and the linear acceleration of the sensor. To avoid this, the $\Delta \mathbf{q}_{acc}$ is interpolated with the identity quaternion, using a gain parameter k_{acc} . Valenti *et. al.* proposes to use LERP interpolation for relatively small corrections. The metric given for this selection is the cosine of the subtended angle Ω between interpolated quaternions. The suggested threshold over which to use *LERP* is $\cos(\Omega) > 0.9$, below that *SLERP* should be used. When interpolating with unit quaternion, the $\cos(\Omega)$ is equal to the scalar part of the $\Delta \mathbf{q}_{acc}$ quaternion. The final correction term becomes:

$$\frac{E}{W} \widehat{\Delta \mathbf{q}}_{acc} = \begin{cases} \frac{LERP(\mathbf{q}_I, \Delta \mathbf{q}_{acc}, k_{acc})}{\|LERP(\mathbf{q}_I, \Delta \mathbf{q}_{acc}, k_{acc})\|} & , \Delta q_{0,acc} > 0.9 \\ SLERP(\mathbf{q}_I, \Delta \mathbf{q}_{acc}, k_{acc}) & , \Delta q_{0,acc} \leq 0.9 \end{cases} \quad (4.11)$$

4.3. Madgwick filter

4.4. Extended Kalman Filter

5. Experimentation

5.1. Equipment

5.2. Methodology

5.3. Results

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